Markscheme

Mathematics: analysis and approaches
Practice question bank

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119. (a)

5 people, including A, B and C, are to be seated in a row. Person C must sit at one of the two ends, and A and B must not sit next to each other.

Find the number of valid seating arrangements.

C at an end: 2 choices; remaining 4 people fill the rest: $2 \times 4! = 48$ arrangements *(M1)* **A1**

among these, count arrangements with A, B adjacent (treat as a block): $2 \times (2 \times 3!) = 24$ *(M1)* **A1**

subtract: $48 - 24$ *(M1)*

$= 24$ **A1**

[7 marks]